

³The world map is based on Natural Earth's world map with the ISO standard point of view for disputed territories, which means that the map may not contain some disputed territories. We note that in addition to the countries shown in gray ("Claude not available"), we do not operate in the Ukrainian regions Crimea, Donetsk, Kherson, Luhansk, and Zaporizhzhia. In accordance with international sanctions and our commitment to supporting Ukraine's territorial integrity, our services are not available in areas under Russian occupation.

⁴"No data" applies to countries with partially missing data. Some territories (e.g., Western Sahara, French Guiana) have their own ISO-3611 code. Some of these have some usage, others have none. Since the Anthropogenic AI Usage Index is calculated per working-age capita based on working age population data from the World Bank, and population data is not readily available for all of these territories, we cannot calculate the AUI for these territories.

⁵ We exclude the Seychelles from all geographic analyses because a large fraction of usage we saw during the sampling dates was abusive traffic.

⁶ We exclude Wyoming from all US state analyses because a large fraction of usage we saw during the sampling dates was abusive traffic.

CHAPTER 4

Tasks and productivity

In this chapter, we examine how time savings, success rates, and autonomy vary across task types, and what this entails for potential impacts on jobs and productivity.

The patterns reveal that more complex tasks yield greater time savings, but that this trades off against reliability. In a simple task removal exercise inspired by [Autor and Thompson \(2025\)](#), Claude’s tendency to cover higher-education tasks produces a net deskilling effect across most occupations, as the tasks AI handles are often the more skilled components of a job.

Claude usage spans a meaningful fraction of tasks across a growing share of occupations. We incorporate success rates into a richer model of job coverage; some occupations with modest coverage see large effects because AI succeeds on their most time-intensive work. Adjusting productivity estimates for task reliability roughly halves the implied gains, from 1.8 to about 1.0 percentage points of annual labor productivity growth over the next decade. However, these estimates reflect current model capabilities, and all signs suggest that reliability over increasingly long-running tasks will improve.

Tradeoffs in task acceleration

Our estimates suggest that, in general, the more complex tasks in our data yield a greater time savings (or “speedup”) from AI. We derive this by having Claude estimate both how long a task would take a human working alone and the duration when human and AI work together, which we validated in [previous work](#). Speedup is then the human-alone time divided by the human-with-AI time. So reducing a 1 hour task to 10 minutes would give a 6x speedup.

The left panel of Figure 4.1 below gives the average speedup against our core measure of task complexity, the human years of schooling required to understand the inputs, all at the O*NET task level.¹ It shows that in Claude.ai conversations, for example, prompts requiring 12 years of schooling (a high school education) enjoy a speedup of 9x, while those requiring 16 years

of schooling (a college degree) attain a 12x speedup. This implies that productivity gains are more pronounced for use cases requiring higher human capital, consistent with evidence that white collar workers are far more likely to adopt AI (e.g., [Bick et al 2025](#)).

Throughout the range of task complexity, the speedup is higher for API users. This could reflect the nature of the API data, which is restricted to single-turn interactions, and that API tasks have been specifically selected for automation.

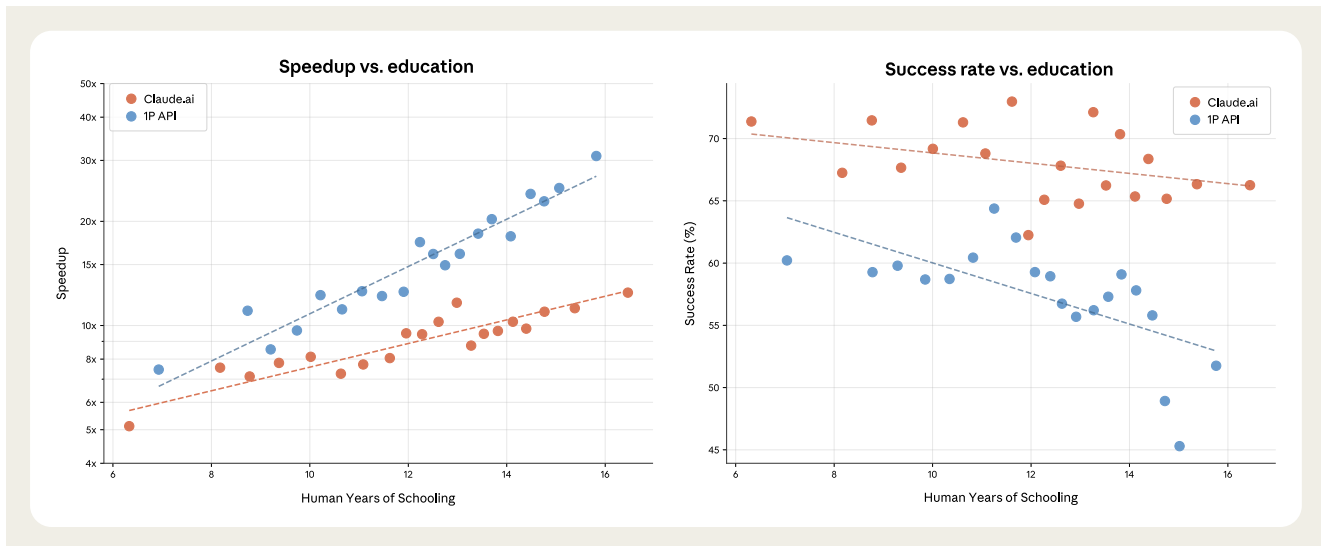


Figure 4.1: Speed up (panel a) and Success rate (panel b) vs. Human years of schooling.

The panel on the left shows a binned scatterplot of the bivariate relationship between speedup and human years of schooling, all measured at the O*NET task level and split by platform. The dashed lines show the fit from a linear regression. The panel on the right shows the same relationship with the success rate in the y-axis.

The results also capture a tradeoff, however. More complex tasks have a lower task success rate, as shown in the panel on the right. On Claude.ai, for example, tasks requiring less than a high school education (e.g., answering basic questions about products) attain a 70% success rate, but this drops to 66% for college-level conversations like developing analysis plans. Still, accounting for the difference in success rates—by either excluding low-success tasks or discounting speedups by success probability—does not eliminate the education gradient: complex tasks still show greater net productivity gains.

One way to examine the implications of the education gradient is to look at the share of automation across the education levels required to understand the inputs. If high-education tasks show relatively more automation, it could signal more exposure for white collar workers. Here, though, the message is unclear:

the automation share is essentially unrelated to the human levels of education required to write the prompt (Appendix Figure A.1).² On both Claude.ai and 1P API, tasks across education levels show automation patterns in roughly equal shares.

In what contexts do users defer more to Claude? Claude.ai users give the AI slightly more autonomy when working on more complex tasks. In contrast, API usage shows uniformly lower autonomy at all levels of complexity.

AI autonomy vs human education by platform

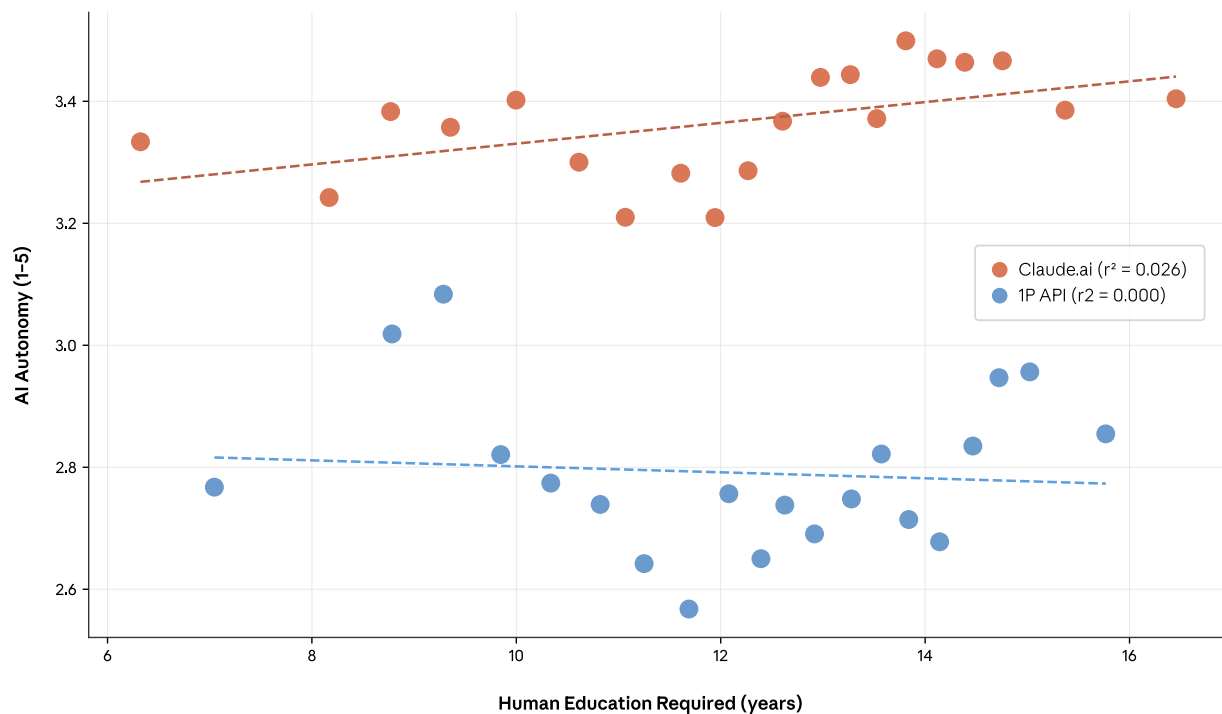


Figure 4.2: AI autonomy vs. human education.

The plot shows a binned scatterplot of the bivariate relationship between AI autonomy and human education required, all measured at the O*NET task level. The dashed lines show the fit from a linear regression.